

Problem

The positive integers $a_1, a_2, \dots, a_n$ are all less than 1000, and

$$\text{lcm}(a_i, a_j) > 1000$$

whenever $i \neq j$. Prove that

$$\sum_{i=1}^n \frac{1}{a_i} < 2.$$

Source: 1951 Russian Mathematics Olympiad

§1 Counting Multiples Hidden Below One Thousand

The condition

$$\text{lcm}(a_i, a_j) > 1000 \quad (i \neq j)$$

implies that no positive integer not exceeding 1000 can be a common multiple of two distinct members of the set $\{a_1, \dots, a_n\}$.

Indeed, if some integer $t \leq 1000$ were divisible by both a_i and a_j , then

$$\text{lcm}(a_i, a_j) \mid t,$$

which would force

$$\text{lcm}(a_i, a_j) \leq t \leq 1000,$$

contrary to the hypothesis.

Thus the sets of multiples of the various a_i below 1000 are pairwise disjoint.

§2 Partitioning the Integers

For each positive integer m , consider the interval

$$\left(\frac{1000}{m+1}, \frac{1000}{m} \right].$$

Let k_m denote the number of the integers a_i lying in this interval.

If

$$\frac{1000}{m+1} < a \leq \frac{1000}{m},$$

then the integers

$$a, 2a, \dots, ma$$

are all at most 1000.

Hence every such a contributes at least m distinct multiples not exceeding 1000.

Since the collections of multiples arising from different a_i are disjoint, the total number of such multiples cannot exceed 1000. Therefore

$$k_1 + 2k_2 + 3k_3 + \dots < 1000.$$

§3 Transforming the Harmonic Sum

For every a_i belonging to the interval

$$\left(\frac{1000}{m+1}, \frac{1000}{m} \right],$$

we have

$$\frac{1}{a_i} \leq \frac{m+1}{1000}.$$

Consequently,

$$\sum_{i=1}^n \frac{1}{a_i} \leq \frac{2k_1 + 3k_2 + 4k_3 + \cdots}{1000}.$$

Observe that

$$2k_1 + 3k_2 + 4k_3 + \cdots = (k_1 + 2k_2 + 3k_3 + \cdots) + (k_1 + k_2 + k_3 + \cdots).$$

Since

$$k_1 + k_2 + k_3 + \cdots = n,$$

it follows that

$$2k_1 + 3k_2 + 4k_3 + \cdots < 1000 + n.$$

Therefore

$$\sum_{i=1}^n \frac{1}{a_i} < \frac{1000 + n}{1000}.$$

It remains to show that $n < 1000$.

This is immediate because the integers a_i are distinct positive integers less than 1000, so at most 999 such integers can exist. Hence

$$1000 + n < 2000,$$

and therefore

$$\sum_{i=1}^n \frac{1}{a_i} < \frac{2000}{1000} = 2.$$

Thus

$$\boxed{\sum_{i=1}^n \frac{1}{a_i} < 2}.$$

§4 A Sharper Observation

The bound may be improved. Consider the greatest odd divisor of each a_i .

If two numbers a_i and a_j had the same greatest odd divisor, then one would differ from the other by multiplication by a power of 2. Consequently one would divide the other, giving

$$\text{lcm}(a_i, a_j) = \max(a_i, a_j) \leq 1000,$$

contradicting the hypothesis.

Hence the greatest odd divisors of the a_i are all distinct. Since there are only 500 odd integers not exceeding 1000, we obtain

$$n \leq 500.$$

Substituting this into the preceding estimate yields

$$\sum_{i=1}^n \frac{1}{a_i} < \frac{1000 + 500}{1000} = \frac{3}{2}.$$

Therefore, in fact,

$$\boxed{\sum_{i=1}^n \frac{1}{a_i} < \frac{3}{2}}.$$